Ananya Nandy

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Education

University of California, Berkeley — *Berkeley, CA*
Ph.D. in Mechanical Engineering — *August 2019 – Exp. 2024*

- Recipient of Departmental First-Year Fellowship

Massachusetts Institute of Technology (MIT) — *Cambridge, MA*
B.S. in Mechanical Engineering (GPA: 4.9/5.0) — *September 2015 – June 2019*

- Thesis: Fabrication Process and Electromechanical Characterization of a Sliding Connector System for Modular Stretchable Electronics
- Tau Beta Pi Engineering Honor Society, Pi Tau Sigma Mechanical Engineering Honor Society

Research Experience

University of California, Berkeley — *Berkeley, CA*
Ph.D. Student — *August 2019 – Present*

- Research assistant in the group of Professor Kosa Goucher-Lambert.
- Investigating functional modeling for human-centered computation in engineering design.

Busch Vacuum Pumps and Systems, R&D Dept. — *Maulburg, Baden-Württemburg, Germany*
Research Intern — *June 2019 – August 2019*

- Designed and executed experiment to characterize acoustics of claw compressor.
- Developed MATLAB tool to determine acoustic impedance.
- Incorporated acoustic impedance to improve performance of silencer acoustic simulation.
- Presented results to company founder, managing partner, and global R&D manager.

MIT Media Lab, Responsive Environments Group (Prof. Joseph Paradiso) — *Cambridge, MA*
Undergraduate Researcher — *September 2018 – May 2019*

- Manufactured novel electro-mechanically robust connector systems to connect modules.
- Conducted electrical and mechanical characterization tests on connectors for stretchable electronic sensor module (part of a modular wearable).
- Wrote senior thesis on results.

Mitsubishi Electric R&D Center, Solution Engineering Dept. — *Amagasaki, Hyogo, Japan*
Smart Systems Group Research Intern — *June 2017 – August 2017*

- Applied machine learning methods in Python to analyze energy data for consumer and supplier energy management within smart systems.
- Successfully disaggregated appliance level data from smart meter data.

Work Experience

Sistine Solar, Greentown Labs — *Somerville, MA*
Mechanical Engineering Intern — *June 2018 – August 2018*

- Headed product development process for device to efficiently apply thin SolarSkin films to solar panels to improve solar panel aesthetics.
- Executed design, rapid prototyping, material procurement, manufacturing, and testing for first prototype from concept to completion.
- Worked within a small, fast-paced team and delivered product with a plan for manufacturing at scale.

Publications and Presentations

Poster

Nandy, A. "Effect of Figure Skating Jump Entry on Jump Quality", MIT Sports Technology Summit. Cambridge, 2018

Extracurricular Activities and Outreach

Science Policy Group at Berkeley *September 2019 - Present*

- Engaging in science communication through op-eds (currently: plastics and sustainable products).

MIT Maker Workshop Mentor *September 2018 – May 2019*

- Trained students on waterjet and maintained machine.
- Supervised and mentored users at student-run makerspace.

Lab Assistant for Design and Manufacturing II *February 2019 – May 2019*

- Assisted lab section of 10 students with CAD, CAM, and manufacturing for mass-manufactured yo-yo project.

MIT Global Teaching Labs *January 2018*

- Developed hands-on Arduino workshops to introduce engineering to high school students.
- Taught physics and the Arduino workshops to over 200 3rd, 4th, and 5th year students at Leonardo da Vinci Scientific High School in Milan, Italy.

MIT Design for America Project Director *May 2017 – May 2018*

- Led 5 teams through yearlong user-centered design projects with the topics of reuse on-campus, indoor air pollution, storage for the homeless, snow removal in Boston, and women in STEM.
- Developed timelines/design process lessons and organized semester design reviews.

Captain of MIT Figure Skating Team *August 2016 – May 2018*

- Coordinated logistics for team of 5 – 10 for intercollegiate competitions.
- Organized team social events and developed team spirit activities.
- Assisted with weekly group lessons for >50 members.

Peer Mentor at Maseeh Hall *August 2016 – May 2017*

- Advised 20 first-year MIT students in residence hall.
- Implemented events and communicated resources to ensure a smooth first-year transition.
- Mediated conflicts and problems for students.

Skills

SolidWorks, Autodesk Fusion 360, MATLAB, Python (Anaconda & Keras), R, CNC/Machining (Mill & Lathe), Waterjet, Laser Cutter, 3D Printing, Electronics/Arduino, Wet Lab Experimentation, Instron Tensile Testing, Microsoft Office

Coursework

User Interface Design, Designing for Emerging Technologies, Introduction to Robotics, Neural Control of Movement, Machine Learning, Mechanics/Materials, Dynamics/Controls, Thermal Fluids Engineering, Measurement/Instrumentation, Design/ Manufacturing, Design for the Developing World, Principles and Practices of Assistive Technology, Electronics for Mechanical Systems, Numerical Computation, Differential Equations, Multivariable Calculus, Physics (Mechanics / Electricity & Magnetism)